What does “Advanced Classical Mechanics” mean?

1. **Mechanics:** Study of motions of objects in **space** as a function of **time**.

2. **Classical:** Study of object motions using mostly Lagrangian Mechanics, Hamiltonian Mechanics, or Newtonian Mechanics (using Newton's laws of motion):

    1. A body remains at rest, or in motion at a constant speed in a straight line, unless it is acted upon by a force.
    2. At any instant of time, the net force on a body is equal to the body's acceleration multiplied by its mass or.
    3. If two bodies exert forces on each other, these forces have the same magnitude but opposite directions.

    These formulations mechanics pertain motions of objects that are not too massive, too small, or moving too fast, compared to objects we see in daily life.
    When we approach the boundaries of classical mechanics mentioned above, we begin to enter the realm of physics that is more accurately described by other theories, such as

    1. relativistic mechanics which pertains objects that move too fast or are too heavy,
    2. Quantum mechanics which pertains objects that are too small.

3. **Advanced:** Study of complex or perhaps chaotic systems that go well beyond simply applying the Newton’s laws of motion.

What do space and time mean?

- Physical space and time are the continuum within which objects move.

- Although objects are free to move in different directions in space within the realm of classical mechanics, they can only move in one direction (forward) along the time axis (there are exceptions to this notion when we exit the realm of classical mechanics to enter relativistic or quantum mechanics).

- Mathematically speaking, **space** is a set (collection of elements or points) with a structure that defines the relationship among the elements. As such, time is not inherently different from space mathematically speaking.

- While there are many types of spaces in mathematics, the most relevant type of spaces for physicists are known as **geometric spaces**.

- Within the family of **geometric spaces**, the **Euclidean geometry** (named after ancient Greek mathematician **Euclid**)

The Euclidean geometry **axioms** (statments that are accepted as truth without proof):

1. For any two different points,
   (a) there exists a line containing these two points, and
   (b) this line is unique.

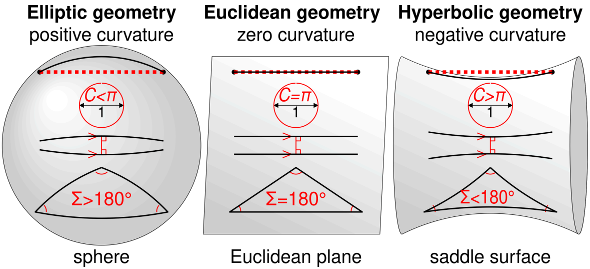
2. A straight line segment can be prolonged indefinitely.

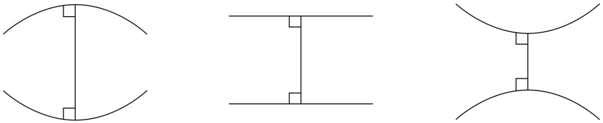
3. A circle can be constructed when a point for its center and a distance for its radius are given.

4. All right angles are equal.

5. For any line L and point p not on L,
  (a) there exists a line through p not meeting L, and
  (b) this line is unique.

While Euclidean geometry is most widely used in classical mechanics, it is not the only one:

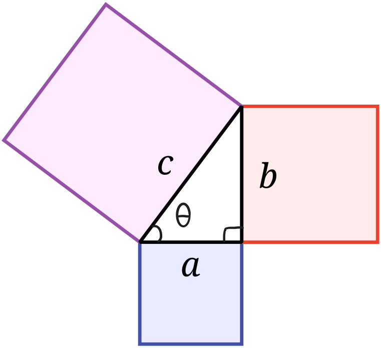




Question 1: A fundamental relation in **Eudlidean geometry** is the Pythagorean theorem, which states that the following relationship holds among the sides of a **right triangle**:

$$a^2 + b^2 = c^2$$

Prove this relationship.



Consider again the above right triangle. In many physical and mathematical calculations we encounter the ratio of the triangle sides to the hypotenuse, so much that we prefer to give them special names. **The trigonoetric functions (sine, cosine, tangent, and cotangent)**.

$$\sin(\theta) = \frac{b}{c} \; ; \; \cos(\theta) = \frac{a}{c}$$

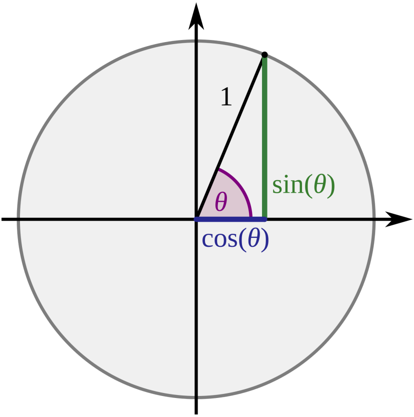
$$\tan(\theta) = \frac{b}{a} \; ; \; \cot(\theta) = \frac{a}{b}$$

Question 2: What are the possible ranges of values of the above trigonometric functions?

Question 3: Using the above definitions and the Pythagorean theorem, prove the following trigonometric relationship holds.

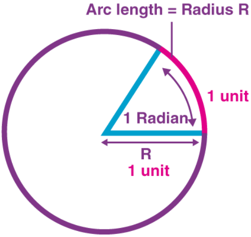
$$\sin^2(\theta) + \cos^2(\theta) = 1$$

With the above proof, you may have realized that we can visualize the trigonometric functions in a unit-radius circle as in the following:



How do we measure an angle?

While there are several historical units for measuring angles (the most popular being 360 degrees for a full circle), the scientific community unanimously prefers the angle unit defined by the **International System of Units (SI)**, named **radian**, and defined as the angle at the center of a circle in a plane that subtends an arc whose length equals the radius of the circle.



Question 4: Using the above definitions, show the following relationships hold (Note that a negative angle implies an angle in the clockwise direction on the circle above).

$\sin(-\theta) = -\sin\theta$	$\sin\left(\frac{\pi}{2} - \theta\right) = \cos\theta$	$\sin(\pi - \theta) = +\sin\theta$
$\cos(-\theta) = +\cos\theta$	$\cos\left(\frac{\pi}{2} - \theta\right) = \sin\theta$	$\cos(\pi - \theta) = -\cos\theta$

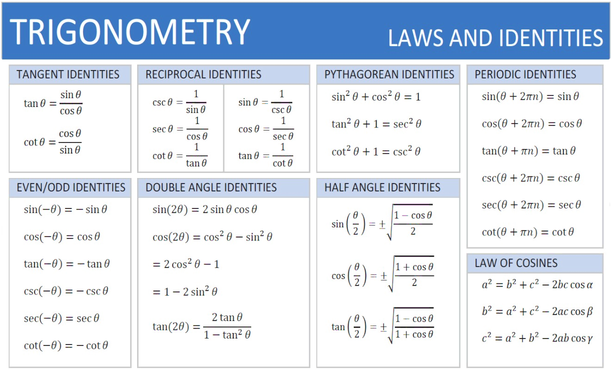
$\sin\left(\frac{3\pi}{2}-\theta\right)=-\cos\theta$	$\sin(2\pi-\theta)=-\sin(\theta)=\sin(-\theta)$
$\cos\left(\frac{3\pi}{2}-\theta\right)=-\sin\theta$	$\cos(2\pi-\theta)=+\cos(\theta)=\cos(-\theta)$

Question 5: Prove the following identities hold (**Hint**: Draw two right triangles on top of each other such the sum or subtraction of their angles at their joint corner forms the resulting angles on the left hand side of the following identities).

$$\begin{aligned}
\sin(\alpha+\beta)&=\sin\alpha\cos\beta+\cos\alpha\sin\beta\\
\sin(\alpha-\beta)&=\sin\alpha\cos\beta-\cos\alpha\sin\beta\\
\cos(\alpha+\beta)&=\cos\alpha\cos\beta-\sin\alpha\sin\beta\\
\cos(\alpha-\beta)&=\cos\alpha\cos\beta+\sin\alpha\sin\beta
\end{aligned}$$

$$\tan(\alpha\pm\beta)=\frac{\tan\alpha\pm\tan\beta}{1\mp\tan\alpha\tan\beta}$$

Without any further proofs, we consider and use the following trigonometric identities as needed in this course.



TRIGONOMETRY LAWS AND IDENTITIES

TANGENT IDENTITIES

$\tan\theta=\frac{\sin\theta}{\cos\theta}$

$\cot\theta=\frac{\cos\theta}{\sin\theta}$

RECIPROCAL IDENTITIES

$\csc\theta=\frac{1}{\sin\theta}$ $\quad\sin\theta=\frac{1}{\csc\theta}$

$\sec\theta=\frac{1}{\cos\theta}$ $\quad\cos\theta=\frac{1}{\sec\theta}$

$\cot\theta=\frac{1}{\tan\theta}$ $\quad\tan\theta=\frac{1}{\cot\theta}$

PYTHAGOREAN IDENTITIES

$\sin^2\theta+\cos^2\theta=1$

$\tan^2\theta+1=\sec^2\theta$

$\cot^2\theta+1=\csc^2\theta$

PERIODIC IDENTITIES

$\sin(\theta+2\pi n)=\sin\theta$

$\cos(\theta+2\pi n)=\cos\theta$

$\tan(\theta+\pi n)=\tan\theta$

$\csc(\theta+2\pi n)=\csc\theta$

$\sec(\theta+2\pi n)=\sec\theta$

$\cot(\theta+\pi n)=\cot\theta$

EVEN/ODD IDENTITIES

$\sin(-\theta)=-\sin\theta$

$\cos(-\theta)=\cos\theta$

$\tan(-\theta)=-\tan\theta$

$\csc(-\theta)=-\csc\theta$

$\sec(-\theta)=\sec\theta$

$\cot(-\theta)=-\cot\theta$

DOUBLE ANGLE IDENTITIES

$\sin(2\theta)=2\sin\theta\cos\theta$

$\cos(2\theta)=\cos^2\theta-\sin^2\theta$

$=2\cos^2\theta-1$

$=1-2\sin^2\theta$

$\tan(2\theta)=\frac{2\tan\theta}{1-\tan^2\theta}$

HALF ANGLE IDENTITIES

$\sin\left(\frac{\theta}{2}\right)=\pm\sqrt{\frac{1-\cos\theta}{2}}$

$\cos\left(\frac{\theta}{2}\right)=\pm\sqrt{\frac{1+\cos\theta}{2}}$

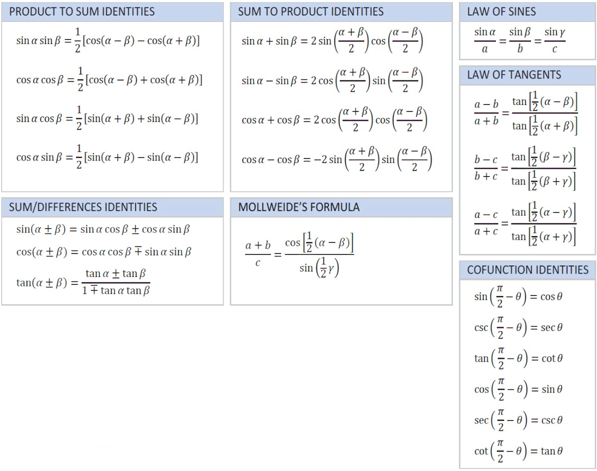
$\tan\left(\frac{\theta}{2}\right)=\pm\sqrt{\frac{1-\cos\theta}{1+\cos\theta}}$

LAW OF COSINES

$a^2=b^2+c^2-2bc\cos\alpha$

$b^2=a^2+c^2-2ac\cos\beta$

$c^2=a^2+b^2-2ab\cos\gamma$



PRODUCT TO SUM IDENTITIES

$$\sin\alpha\sin\beta = \frac{1}{2}[\cos(\alpha-\beta)-\cos(\alpha+\beta)]$$

$$\cos\alpha\cos\beta = \frac{1}{2}[\cos(\alpha-\beta)+\cos(\alpha+\beta)]$$

$$\sin\alpha\cos\beta = \frac{1}{2}[\sin(\alpha+\beta)+\sin(\alpha-\beta)]$$

$$\cos\alpha\sin\beta = \frac{1}{2}[\sin(\alpha+\beta)-\sin(\alpha-\beta)]$$

SUM TO PRODUCT IDENTITIES

$$\sin\alpha+\sin\beta = 2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)$$

$$\sin\alpha-\sin\beta = 2\cos\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right)$$

$$\cos\alpha+\cos\beta = 2\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)$$

$$\cos\alpha-\cos\beta = -2\sin\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right)$$

LAW OF SINES

$$\frac{\sin\alpha}{a} = \frac{\sin\beta}{b} = \frac{\sin\gamma}{c}$$

LAW OF TANGENTS

$$\frac{a-b}{a+b} = \frac{\tan\left[\frac{1}{2}(\alpha-\beta)\right]}{\tan\left[\frac{1}{2}(\alpha+\beta)\right]}$$

$$\frac{b-c}{b+c} = \frac{\tan\left[\frac{1}{2}(\beta-\gamma)\right]}{\tan\left[\frac{1}{2}(\beta+\gamma)\right]}$$

$$\frac{a-c}{a+c} = \frac{\tan\left[\frac{1}{2}(\alpha-\gamma)\right]}{\tan\left[\frac{1}{2}(\alpha+\gamma)\right]}$$

SUM/DIFFERENCES IDENTITIES

$$\sin(\alpha\pm\beta) = \sin\alpha\cos\beta\pm\cos\alpha\sin\beta$$

$$\cos(\alpha\pm\beta) = \cos\alpha\cos\beta\mp\sin\alpha\sin\beta$$

$$\tan(\alpha\pm\beta) = \frac{\tan\alpha\pm\tan\beta}{1\mp\tan\alpha\tan\beta}$$

MOLLWEIDE'S FORMULA

$$\frac{a+b}{c} = \frac{\cos\left[\frac{1}{2}(\alpha-\beta)\right]}{\sin\left(\frac{1}{2}\gamma\right)}$$

COFUNCTION IDENTITIES

$$\sin\left(\frac{\pi}{2}-\theta\right) = \cos\theta$$

$$\csc\left(\frac{\pi}{2}-\theta\right) = \sec\theta$$

$$\tan\left(\frac{\pi}{2}-\theta\right) = \cot\theta$$

$$\cos\left(\frac{\pi}{2}-\theta\right) = \sin\theta$$

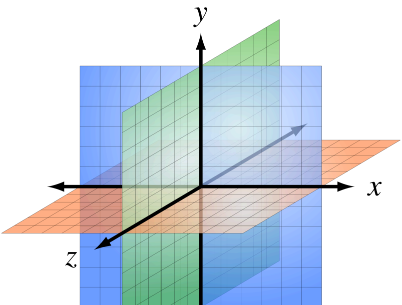
$$\sec\left(\frac{\pi}{2}-\theta\right) = \csc\theta$$

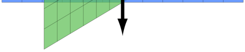
$$\cot\left(\frac{\pi}{2}-\theta\right) = \tan\theta$$

Coordinate systems.

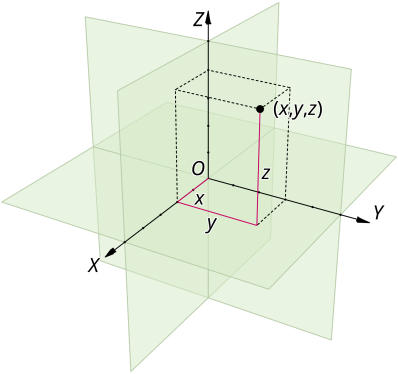
A coordinate system is a collection of numbers that are **uniquely** assigned to elements (typically points) in a given geomtery. These coordinate numbers **uniquely** determine the **position** of different elements in a given geomtery. Within each coordinate system, a reference number (commonly all zeros) is mapped to a reference point within the geometry call the **origin of the coordinate system**.

For example, in Euclidean geometry, the Cartesian coordinates is the most natural and popular choice comprised of N perpendicular planes in N dimensions as illustrated below for three dimensions.





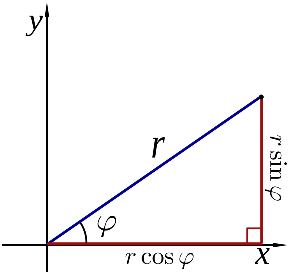
The three-dimensional Euclidean space



The three-dimensional Cartesian coordinates with an example position x, y, z specified for a point.

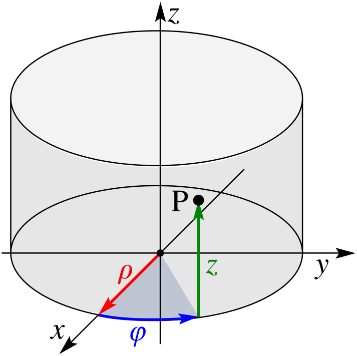
Are there any alternatives to Cartesian coordinates system?
Yes. For example, the **polar coordinates** in two dimensions where points are **uniquely** specified from the coordinates' origin with an angle and distance from origin.

Question 6: Write down the set of equations that can convert a Cartesian coordinate (x, y) to Polar coordinate as shown below.



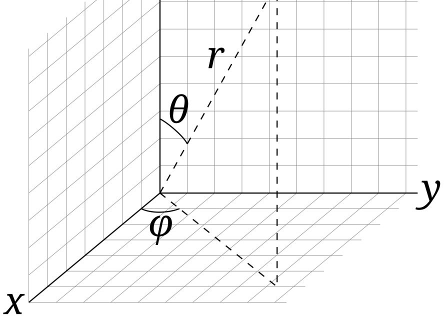
Are there other alternatives to Cartesian and Polar coordinates systems?

Yes. For example, the Cylindrical coordinate system in three-dimensions. We can consider the Cylindrical coordinate system as an extension of the Polar coordinates from two dimensions to three dimensions.



Are there yet other alternatives to Cartesian, Polar, and Cylindrical coordinates systems?

Yes. For example, the **Spherical coordinate system** in three-dimensions is defined by two (polar and azimuthal) angles and a radial distance from the coordinate system's origin. The Spherical coordinate system has widespread use in geography, astronomy, and many disciplines of physics such as Classical Mechanics, Quantum Physics, and Electrodynamics.



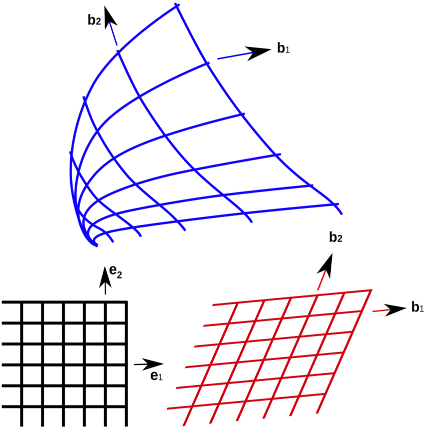
Question 7. Write down the set of equations that convert a Cartesian coordinate (x, y, z) to the Spherical triplet shown above.

Question 8. Write down the set of equations that convert a Cylindrical coordinate triplet to a Spherical triplet shown above.

Are there yet any other coordinate systems?
Yes. There are infinitely many more coordinate systems, but the above mentioned systems are among the most useful.

The usefulness of a coordinate system and its ability to make calculations easier is the primary factor in choosing a specific coordinate system for a specific problem.

Regardless of the choice, all coordinate systems uniquely map each point in the geometric space to a unique collection of numbers.

Question 9. Which one of the following illustrations represent a valid coordinate system.